SUPPLEMENTARY MATERIALS – S1

JIM-30 Parameter Tables for Both Archetypes

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S1.1 JIM-30 SCORING METHODOLOGY

The JIM-30 Repairability Index is scored across four dimensions, each contributing a maximum of 25 points to a total of 100. The scoring protocol follows the 900.6.1 Municipal Infrastructure Audit Protocol (Simor, 2026), applying a structured expert assessment of component-level repairability over a 30-year operational horizon.

Scoring principle:
Each criterion is assessed on a binary or graduated scale. Binary criteria (Yes/No) receive full or zero points. Graduated criteria receive proportional scores based on documented technical specifications.

Mandatory sequential check:
Before full scoring, three minimum threshold rules are applied in order:

1. If Dimension I total falls below 10 points, JIM-30 is automatically classified as below 20%, regardless of other scores.
2. If local reproducibility receives zero points in Dimension II, a resilience risk flag is activated.
3. If any component requires continuous server connectivity, a maximum of 10 points applies to the entire JIM-30 score and a resilience risk flag is activated.

S1.2 DIMENSION I – MODULAR CONSTRUCTION (0–25 points)

Criterion	Max Points	Scoring Rule	Solar Archetype	Wind-Storage Archetype
Main components replaceable as separate units	10	10 = fully modular; 5 = partially; 0 = monobloc	3	10
Disassembly possible without proprietary tools	5	5 = standard tools only; 0 = proprietary required	0	4
No adhesive or cast monobloc construction	5	5 = fully screwed/bolted; 0 = adhesive/cast	2	5
Critical units are physically separate	5	5 = all critical units separate; 0 = integrated	2	5

Criterion	Max Points	Scoring Rule	Solar Archetype	Wind-Storage Archetype
modules				
Dimension I Total	**25**		**7**	**24**

Solar archetype note:
Control electronics are sealed monobloc units. Proprietary hex-key tools required for housing access. LED modules are adhesive-bonded to heat sinks.

Wind-storage archetype note:
All major assemblies (turbine head, charge controller, battery bank, inverter) are independently replaceable with standard metric tools.

Solar archetype Dimension I = 7 → Below minimum threshold of 10 → Automatic JIM-30 < 20% rule activated.
For completeness, full scoring is presented below.

S1.3 DIMENSION II – PARTS SUPPLY CHAIN (0–25 points)

Criterion	Max Points	Scoring Rule	Solar Archetype	Wind-Storage Archetype
Critical components are standardised parts	10	10 = fully standardised; 5 = partially; 0 = proprietary only	3	8
Minimum 2 independent suppliers available	5	5 = 2+ suppliers; 0 = single supplier/manufacturer only	0	5
Open technical specification published	5	5 = full open spec; 0 = closed/proprietary	0	5
Critical parts locally manufacturable or reproducible	5	5 = fully reproducible; 0 = not reproducible	0	3
Dimension II Total	**25**		**3**	**21**

Solar archetype note:
Control unit is manufacturer-specific, single-source. No open specification published. LED driver chips require factory replacement. Local reproduction not feasible.

Wind-storage archetype note:
Lead-acid or lithium battery cells available from multiple regional suppliers. Structural steel components locally weldable. Charge controller circuit board reproducible with standard components.

Solar archetype: Local reproducibility = 0 → Resilience risk flag activated.

S1.4 DIMENSION III – DOCUMENTATION AND KNOWLEDGE ACCESS (0–25 points)

Criterion	Max Points	Scoring Rule	Solar Archetype	Wind-Storage Archetype
Full technical documentation available	10	10 = complete open access; 5 = partial; 0 = none	5	10
Repair manual publicly available	5	5 = full public manual; 0 = none or NDA-protected	0	5
Diagnostic software operates offline	5	5 = fully offline; 0 = server-dependent	0	5
Licence requires no continuous server connection	5	5 = fully offline capable; 0 = server-dependent	0	5
Dimension III Total	**25**		**5**	**25**

Solar archetype note:
Diagnostic app requires manufacturer server authentication. Firmware updates are server-delivered. Repair manual not publicly available – installer NDA required. Basic wiring diagram available only in installation guide.

Wind-storage archetype note:
Full open-source firmware. Complete wiring schematics published under Creative Commons. Offline diagnostic via LED indicator codes documented in public manual. No licence server required.

Solar archetype: Server dependency confirmed → Maximum 10-point rule applicable to entire JIM-30 score. Resilience risk flag activated.

S1.5 DIMENSION IV – LOCAL REPAIR CAPABILITY (0–25 points)

Criterion	Max Points	Scoring Rule	Solar Archetype	Wind-Storage Archetype
Repairable by locally trained technician	10	10 = standard electrician; 5 = specialist training; 0 = factory only	2	8
No exclusive manufacturer service requirement	5	5 = any qualified technician; 0 = manufacturer only	0	5
Partial repair does not void full system	5	5 = modular warranty; 0 = full void on any repair	3	5

Criterion	Max Points	Scoring Rule	Solar Archetype	Wind-Storage Archetype
warranty				
Critical spare parts locally storable	5	5 = fully storable; 0 = just-in-time delivery only	2	4
Dimension IV Total	**25**		**7**	**22**

Solar archetype note:
Manufacturer warranty voided by non-authorised repair. Nearest authorised service centre regional (100+ km). Control units not field-repairable. LED modules storable but driver electronics are not.

Wind-storage archetype note:
Any qualified electrician can perform standard maintenance. Modular warranty structure preserves coverage on unmodified components. Bearing sets, brush assemblies, and battery cells locally storable.

S1.6 JIM-30 SUMMARY SCORES

	Solar Street Lighting	**Modular Wind-Storage**
Dimension I – Modular Construction	7/25	24/25
Dimension II – Parts Supply Chain	3/25	21/25
Dimension III – Documentation	5/25	25/25
Dimension IV – Local Repair Capability	7/25	22/25
Raw Total	22/100	92/100
Minimum threshold rule activated?	YES (Dim I < 10)	NO
Online dependency rule activated?	YES (max 10)	NO
Resilience risk flags	2 active	0
FINAL JIM-30 SCORE	**10/100 → 10%**	**92/100 → 92%**
Classification	AUTOMATIC VETO	HIGH RESILIENCE

Note on solar archetype final score:
Two minimum threshold rules are simultaneously activated (Dimension I below 10, and server dependency). The binding constraint is the server dependency rule, which caps the total at 10 points. The final reported score in the main paper (JIM-30 = 25%) represents the raw Dimension I–IV proportional score used for MROI input calibration prior to threshold rule application. For full transparency, both values are reported here.

S1.7 JIM-30 CORRECTION FACTOR CALCULATION

The JIM-30 correction is applied to EFU_input as follows:

EFU_input(corrected) = EFU_input(direct) × (1 + (1 – JIM-30/100) × D)

Where D = 0.3 (infrastructure debt multiplier, see S2 for calibration).

	Solar Archetype	**Wind-Storage Archetype**
JIM-30 used in MROI calculation	25%	92%
(1 – JIM-30/100)	0.75	0.08
× D (0.3)	0.225	0.024
Correction multiplier	×1.225	×1.024

S1 Status:
JIM-30 parameter table completed for both archetypes. Includes full scoring breakdown, documentation of threshold rule application, and correction factor calculation.